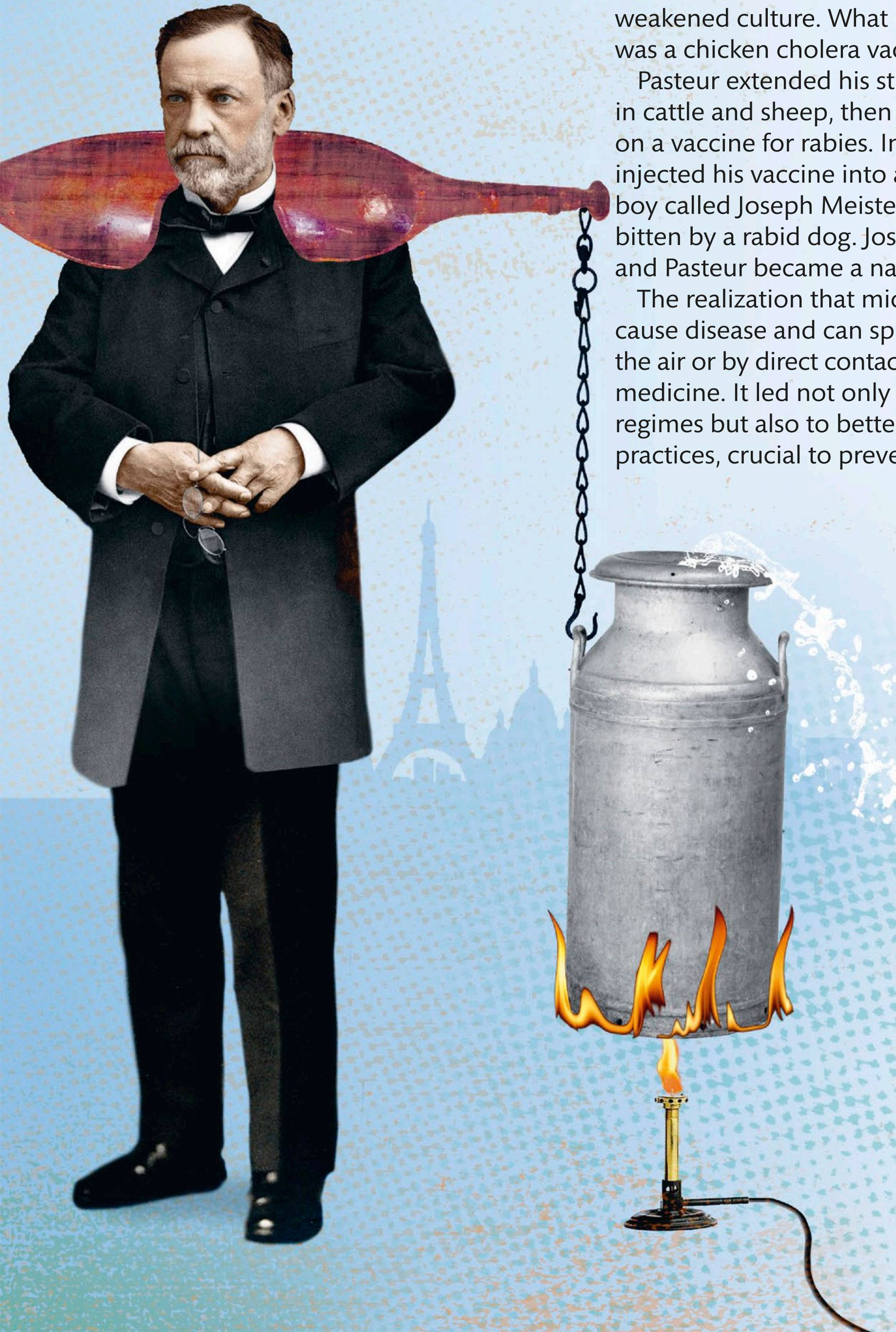
cholera microbes in his laboratory. When inoculated (infected) with these microbes, his chickens became ill and died. But on one occasion, in 1879, the microbe culture was abandoned for a month while Pasteur went on vacation. When Pasteur inoculated the birds with it, they became ill but recovered. He then inoculated them with fresh cultures, which had killed other birds, and they survived. Pasteur realized that his chickens had developed an immunity to cholera as a result of exposure to the weakened culture. What he had created was a chicken cholera vaccine.

Pasteur extended his studies to anthrax in cattle and sheep, then began to work on a vaccine for rabies. In 1885, he injected his vaccine into a 9-year-old boy called Joseph Meister, who had been bitten by a rabid dog. Joseph survived, and Pasteur became a national hero.

The realization that microorganisms cause disease and can spread through the air or by direct contact revolutionized medicine. It led not only to vaccination regimes but also to better basic hygiene practices, crucial to preventing infection.



GAVE **MEISTER 13** VACCINE **INJECTIONS** OVER **10 DAYS; MEISTER RECOVERED**

DEVISED METHODS TO **PROTECT HUMANITY** FROM **2 DEADLY DISEASES—ANTHRAX** AND **RABIES**